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PROFILE

I am a senior student in Engineering and Computer Science at the University of Bologna (Cesena), with my thesis defense scheduled for October 2026. I have experience in full-stack development, open-source scientific simulators, and containerized development environments, with a constant focus on clean architectures and maintainable code. I also program for the sheer joy of it: physical simulations, algorithm visualizers, and mobile apps.

EXPERIENCE

University of Bologna

05/03/2026 — 10/06/2026

Internship ~ Software development

Cesena, Italy

- Designed and developed by Socratic C: a modular, containerized web-based IDE for teaching C, featuring a built-in terminal for compiling and running code.
- Implemented a chat stream featuring a Socratic AI tutor with multi-provider support (cloud and local models via Ollama), designed to guide students without providing direct answers.

Trust-IT Services

23/03/2022 - 22/09/2022

Full Stack Developer

Pisa, Italy

- Porting of the official ECMWF (European Centre for Medium-Range Weather Forecasts) website: migration of PHP modules from Drupal 7 to Drupal 9.
- Front-end and back-end development of an internal data aggregation portal.

EDUCATION

University of Bologna

2021 — Today

Bachelor's Degree in Engineering and Computer Science

Cesena, Italy

- All exams passed; thesis defense in October 2026. Design and implementation of a module for the open-source Alchemist simulator (University of Bologna) to integrate real geospatial data (NetCDF/GRIB, CF conventions) from Copernicus services (CDS/EWDS/ADS) as read-only, pull-based environmental layers, with spatiotemporal interpolation and local cache management.

ITT Blaise Pascal

2016 — 2021

Computer science technical high school - 99/100

Cesena, Italy

PROGETTI

GraphSandbox | Svelte, JavaScript, TailwindCSS

- A web app for generating arbitrary graphs and visualizing classical algorithms (BFS, DFS, Dijkstra, Bellman-Ford) and heuristics in real time.

UniUtils | PHP, JavaScript, TailwindCSS

- Full-stack e-commerce platform for buying and selling college textbooks, including frontend, backend, and relational database modeling.

Customer Churn Classifier | Python, scikit-learn

- Machine learning model on a telecom dataset to predict the monthly churn probability per customer, accompanied by a demo web app.

JVMT | Java

- Digital adaptation of the board game Diamant (group project). Software architecture designed with a strong focus on maintainability, actively applying GoF design patterns (Strategy, Factory, Builder, Template Method) to manage the game logic.

COMPETENZE

Programming languages : Java, Kotlin, Python, C, PHP, JavaScript, TypeScript, Bash

Frontend : Svelte/SvelteKit, HTML, CSS, TailwindCSS

Backend : Node.js, Express.js, REST API

Data & AI : SQL, MySQL, scikit-learn, relational modeling

DevOps & Tools : Windows/Linux/Android Operating Systems, Docker, Docker Compose, Git, Jira

Other : TCP/IP, UML, OOP Design patterns, functional programming, CMS Drupal

Languages : Italian (native speaker), English (B2)